

Player Market Value in Elite European Football

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1 Introduction

The European football transfer market has evolved from a subjective scouting exercise into a global financial industry exceeding €7 billion annually. For this study, we operationalize market value using crowd-sourced estimates from Transfermarkt, a metric widely validated in sports economics as a robust proxy for actual transfer fees (Herm et al., 2014). While foundational research identifies performance metrics like goals as primary drivers (Frick, 2007), more recent scholarship highlights a “Superstar Effect” where non-performance factors like brand visibility magnify valuation (Franck & Nüesch, 2012; Rosen, 1981). However, the market has recently bifurcated into a pre-2020 hyper-inflationary era, termed the “Neymar Effect” (Poli et al., 2017), and a post-2020 pandemic correction (Drewes et al., 2021). This trajectory was violently interrupted by the COVID-19 pandemic, which caused an estimated €8.7 billion revenue shortfall across European football (Deloitte, 2021). Existing models typically treat players as static units nested within single clubs, ignoring the dynamic reality where players transfer between teams and accumulate value from both their portable human capital and the institutional prestige of their current employer. Neglecting this cross-classified structure likely misestimates the true drivers of market value in this volatile era. To address these limitations, this study applies a cross-classified multilevel model to a decade of transfer data (2013–2024), centering the analysis at the 2020 pandemic baseline to quantify the impact of this structural break. We attempt to disentangle individual reputation from

club context by investigate four critical dynamics:

1. The non-linear “Boom and Bust” market trajectory. How has the transfer market evolved relative to the 2020 baseline? Specifically, does the market exhibit a non-linear “Boom and Bust” cycle that linear models miss?
2. The biological constraints of the player career arc. How does market valuation evolve across a player’s biological career, and at what specific age does this value peak?
3. The valuation signal of “Squad Status” via relative playing time. To what degree does “Squad Status”, which is defined by relative playing time, is required predict market value independent of raw performance output?
4. The existence of contextual premiums for club diversity or discounts for domestic talent in a globalized labor market. How does squad internationalization affect player market values, and do domestic players face a valuation penalty compared to foreign players?

2 Method

2.1 Sample and Data

We obtained data from Transfermarkt and UEFA official rankings using purposive sampling to select the top 21 European clubs based on 10-year averaged UEFA coefficient rankings¹. From these clubs we identified players who had a full season from 2013 through 2024 resulting in a sample of 7,023 player-season observations from 2,159 unique players across. After listwise deletion of cases with missing citizenship data (approximately 2% loss), players averaged 26.5 years of age ($SD = 4.6$). Critically, 441 players (20.4%) played for multiple clubs during the study period, justifying our cross-classified modeling approach.

Our primary outcome variable is the natural logarithm of the player’s inflation-adjusted

¹UEFA coefficient rankings are a points-based system used by European soccer’s governing body to rank national associations and clubs based on their performance in European competitions, which determines qualification spots and seeding for tournaments like the Champions League and Europa League.

yearly average market value (in 2024 USD). Predictors are categorized into three distinct domains: (1) Player Characteristics, including age, height, position (Attacker, Midfielder, etc.), and citizenship status relative to the club (homegrown vs. foreign); (2) performance and status, captured by goals, assists, red/yellow cards, and playing time; and (3) club context, specifically the proportion of foreign players on the roster and the club’s UEFA coefficient, which serves as a proxy for recent competitive success and prestige. For further descriptive statistics on each variable see Appendix A, Table1.

2.2 Model Specification

Our data exhibit a cross-classified structure where players (i) are constant units, but club membership (j) changes over time (t). When a player transfers from club j_1 to club j_2 , club-level predictors such as internationalization and UEFA ranking that vary by year, are updated to reflect the new club’s characteristics. Figure 1 visualizes our cross-classified design with an example from three players.

Figure 1. Cross-classified Data Structure

Player C	Ajax	Juventus	Juventus	Bayern
Player B	Dortmund	Dortmund	Man City	Man City
Player A	Real Madrid	Real Madrid	Real Madrid	Real Madrid
	2020	2021	2022	2023

Highlights how players transfer between clubs across seasons.

This structure requires modeling player and club as crossed random effects rather than nested effects. Crucially, because club attributes like internationalization and UEFA ranking vary year-to-year, they are modeled as Level 1 (time-varying) predictors rather than static Level 2 traits.

Level 1: Observation Model (Time-Varying Effects)

At the observation level, the log-transformed market value is modeled as a function of the player-club intercept (π_{0ij}), algebraic time trends, biological aging, and dynamic contextual metrics:

$$\begin{aligned}\log(\text{MV})_{tij} = & \pi_{0ij} \\ & + \pi_{1j}(\text{Year}_t) + \pi_{2j}(\text{Year}_t^2) + \pi_{3j}(\text{Year}_t^3) \\ & + \pi_{4j}(\text{Age}_{it} - \bar{A}) + \pi_{5j}(\text{Age}_{it} - \bar{A})^2 \\ & + \pi_{6j}(\text{Mins}_{Z,it}) \\ & + \pi_{7j}(\text{Foreign}_{tj}) + \pi_{8j}(\text{UEFA}_{tj}) \\ & + \text{perf} + \epsilon_{tij}\end{aligned}$$

$$\begin{aligned}\Pi_{\text{perf}} = & \pi_9(\text{Goals}_{tij}) + \pi_{10}(\text{Assists}_{tij}) \\ & + \pi_{11}(\text{RedCards}_{tij}) + \pi_{12}(\text{YellowCards}_{tij})\end{aligned}$$

Note: MV = Market Value. The subscripts tj for Foreign and UEFA denote that these club attributes vary by time t.

Level 2a: Player Model (Time-Invariant Traits)

The player-club intercept (π_{0ij}) is decomposed into a fixed component based on static player characteristics and a random player effect (u_{0i}):

$$\begin{aligned}\pi_{0ij} = & \beta_{0j} + \gamma_1(\text{Height}_i) + \gamma_2(\text{Citizen}_{ij}) + \text{demog} + u_{0i} \\ \text{demog} = & \sum_{k=1}^{K-1} \gamma_{\text{pos},k}(\text{Position}_{ik}) + \sum_{r=1}^{R-1} \gamma_{\text{reg},r}(\text{Region}_{ir})\end{aligned}$$

Level 2b: Club Model (Contextual Trajectories)

The club baseline (β_{0j}) and the time-varying slope parameters (π_{1j}, π_{2j}) include random effects to allow for heterogeneity in both growth velocity and trajectory curvature:

$$\beta_{0j} = \delta_{00} + v_{0j} \quad (\text{Baseline Prestige})$$

$$\pi_{1j} = \delta_{10} + v_{1j} \quad (\text{Linear Growth Deviation})$$

$$\pi_{2j} = \delta_{20} + v_{2j} \quad (\text{Quadratic Curvature Deviation})$$

$$\pi_3 = \delta_{30} \quad (\text{Fixed Cubic Trend})$$

Variance-Covariance Structure

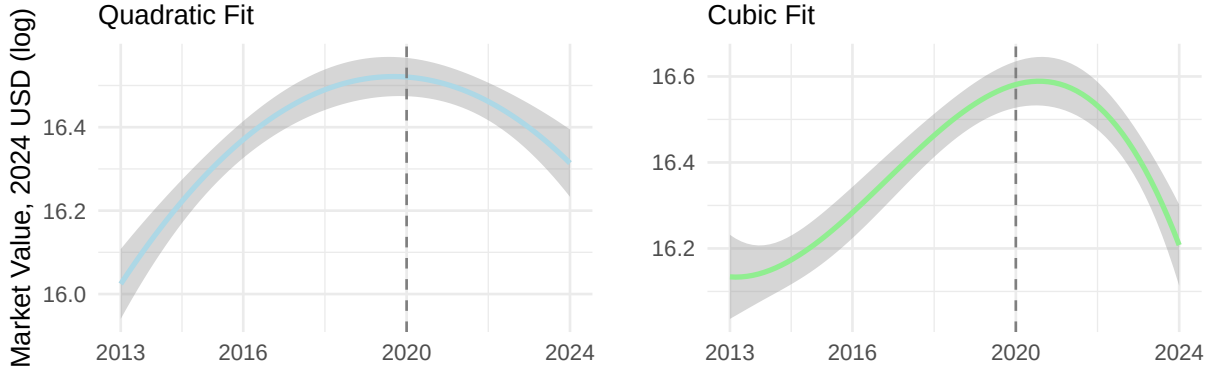
$$u_{0i} \sim N(0, \sigma_u^2) \quad , \quad \begin{pmatrix} v_{0j} \\ v_{1j} \\ v_{2j} \end{pmatrix} \sim N \left(\mathbf{0}, \begin{pmatrix} \sigma_{Int}^2 & \sigma_{01} & \sigma_{02} \\ \sigma_{01} & \sigma_{Lin}^2 & \sigma_{12} \\ \sigma_{02} & \sigma_{12} & \sigma_{Quad}^2 \end{pmatrix} \right)$$

2.3 Variables and Centering Strategies

We apply a strategic centering and transformation strategy to isolate the pandemic’s structural impact and test non-linear market dynamics by centering season year at the onset of COVID-19 in 2020. We include polynomial terms to capture theorized non-linearities (Figure 2): a cubic term for time to model the “Boom-and-Bust” cycle (e.g., pre-pandemic inflation, post-pandemic correction), and a quadratic term for Age to test the biological “Inverted-U” career trajectory. Minutes Played was standardized within each club-year to isolate player squad status (e.g., Starter vs. Bench) from team rotation policies, while performance metrics were position-centered to account for role-specific expectations.²

²Performance expectations vary by role: A goal from a striker is routine while for a defender is exceptional. Position-centering transforms raw counts into “performance above expectation” metrics, ensuring the model captures relative contributions rather than treating goals and assists equally.

Figure 2. Quadratic vs. Cubic Polynomial Fits for Market Value Over Time



Dashed line marks the 2020 pandemic onset.

2.4 Model Building and Statistical Software

We employed sequential model building: (1) null model for ICC estimation; (2) linear time polynomial; (3) cubic time polynomial; (4) random slope for quadratic time at club level³; (5) player/club-season Level 1 predictors with quadratic age, and all code is found in Appendix B. Model selection and comparison was guided by Likelihood Ratio Tests (LRT) to evaluate the statistical significance of nested model improvements, and the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Models yielding significant LRT results ($p < .05$) and lower values on both information criteria were retained for subsequent analysis. We describe each model in Appendix C, Table 1 alongside ICC, AIC, and BIC values.

All analyses were conducted in R version 4.4.1 using the `lme4` 1.1-37 package for mixed-effects model estimation with a Restricted Maximum Likelihood (REML) method. REML was selected to provide unbiased estimates of the variance components by accounting for the degrees of freedom lost when estimating fixed effects. We utilized the BOBYQA optimizer, which handles boundary constraints more effectively than the default Nelder-Mead optimizer,

³We excluded player-level random slopes due to data sparsity; a majority of players appear in the dataset for only one or two seasons, making it mathematically impossible to disentangle their individual growth trajectories from residual error. In contrast, the club-level data provides a robust longitudinal structure with continuous observations across the decade, allowing for the stable estimation of club-specific market value trajectories.

to ensure robust model convergence given the complexity of the random effects structure. Significance tests used Satterthwaite approximations via the `lmerTest` 3.1-3 package, as well as our own function to test random slope variances in Appendix B.

3 Results

The unconditional means (null) model revealed substantial clustering at the player level. Player-level variance accounted for 78% of the total variance ($ICC_{player} = .75$), while club-level variance accounted for 5% ($ICC_{club} = .03$), highlighting that persistent individual heterogeneity dominates market valuations. Regarding temporal dynamics, model comparisons favored a cubic fixed effect for season year over a linear specification, $\chi^2(2) = 45.66, p < .001$. We subsequently tested for club-specific heterogeneity in growth trajectories. A likelihood ratio test using a 50:50 mixture of χ_1^2 and χ_2^2 distributions⁴ confirmed that adding a random linear slope significantly improved model fit compared to the random-intercept model, $\chi^2(5) = 97.13, p < .001$, indicating that clubs differ significantly not only in their growth rates but also in the curvature of their market value trajectories.⁵ Consequently, our final best-fitting model incorporates these complex random effects alongside time-varying player and club predictors ($\chi^2(19) = 5626, p < .001$ compared to random slopes model). Model comparisons are shown in Appendix C, Table 1. Here we output our final model’s variance components and fixed effects.

Our final model converged successfully with no warnings. Q-Q plots confirmed that player-level random intercepts ($n = 2,159$) adhered well to normality, while club-level random effects ($n = 21$) showed minor tail departures expected with few Level 2 units (Maas &

⁴We test whether a random slope with quadratic season year at the club level has better fit to the data. We find evidence supporting the need to include these random slopes using a likelihood ratio test with an alternative hypothesis, $H_A = \tau_{11} > 0$ since variance should be larger than 0. To test this hypothesis we use the likelihood ratio test (LRT) simply by $-2[\log \text{likelihood}_{reduced} - \log \text{likelihood}_{full}]$ to examine the difference. This code can be found in Appendix B.

⁵While we included random linear and quadratic slopes, a random cubic slope was excluded due to over-parameterization. Estimating only 21 clubs led to singularity; therefore, the quadratic specification represents the maximal complexity supported by the data.

Hox, 2004). The residual versus fitted values plot revealed homoscedastic variance with no systematic patterns. Full diagnostic plots are presented in Appendix C.

Table 1: Estimated Variance Components for Final Model

Component	Variance	SD
Player (Reputation)	0.9254	0.9620
Club (Baseline Prestige)	0.0541	0.2326
Club (Linear Growth)	0.0105	0.1025
Club (Quadratic Curvature)	0.0073	0.0855
Residual (Unexplained)	0.2070	0.4550

Table 2: Fixed Effects Estimates from Final Model

Predictor		SE	t	% Change	
Year (linear)	0.002	0.030	0.08	0.2	
Year²	-0.270	0.024	-11.05	-23.7	***
Year³	-0.117	0.011	-10.98	-11.0	***
Height	0.010	0.004	2.81	1.0	**
Defender	-0.403	0.058	-6.97	-33.2	***
Goalkeeper	-1.059	0.095	-11.17	-65.3	***
Midfielder	-0.190	0.058	-3.29	-17.3	**
Domestic citizen	-0.185	0.031	-5.89	-16.9	***
Age (linear)	0.019	0.004	5.02	1.9	***
Age²	-0.024	0.000	-63.08	-2.4	***
goals	0.005	0.002	2.36	0.5	*
assists	0.010	0.003	3.51	1.0	***
Red cards	0.009	0.022	0.42	0.9	
Yellow cards	-0.002	0.003	-0.49	-0.2	
Minutes (Z)	0.350	0.013	26.20	41.9	***
UEFA Coefficient	0.000	0.001	-0.39	0.0	
Proportion of Foreign Players	0.103	0.104	0.99	10.8	

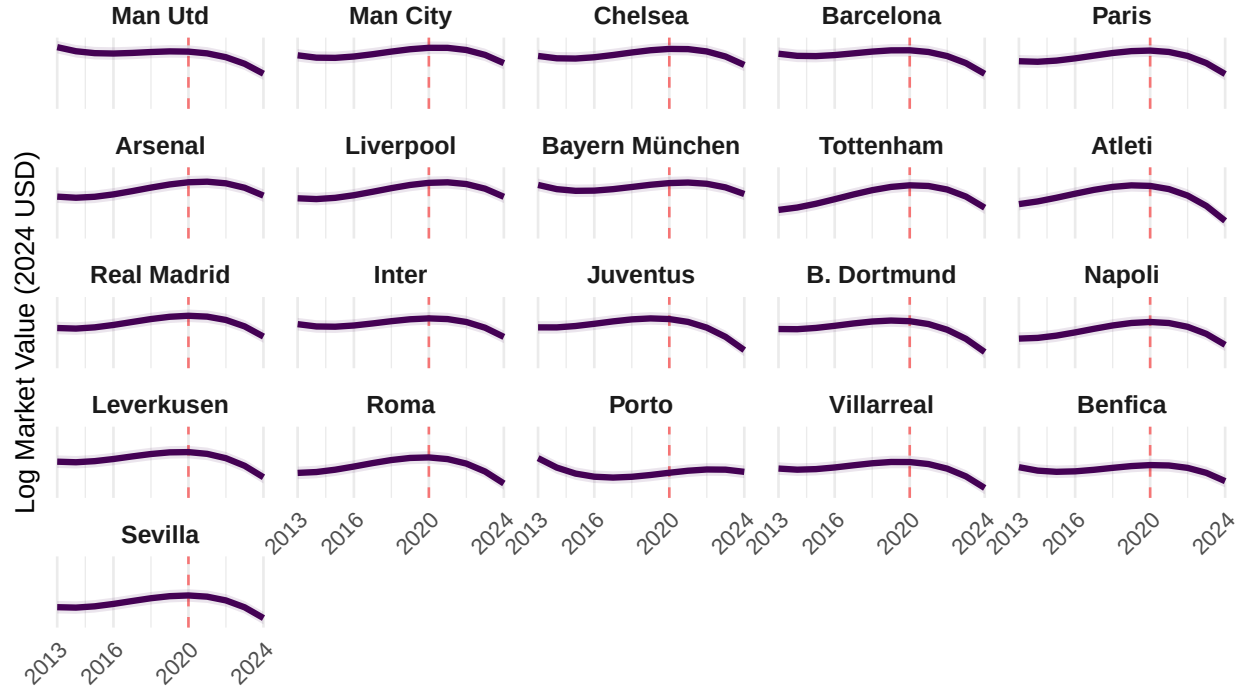
Note: *** $p < .001$, ** $p < .01$, * $p < .05$

3.0.1 Market Evolution

Using raw polynomials centered at 2020 ($t = 0$), our model mathematically pinpoints the moment the market froze. The linear coefficient at the 2020 baseline is effectively zero ($\pi_1 = 0.00, p = 0.93$), indicating that at the exact onset of the pandemic, the relentless inflation of the previous decade came to a complete halt. The significant negative quadratic ($\pi_2 = -0.27$) and cubic ($\pi_3 = -0.12$) terms confirm the transition from a pre-2019 “Boom” into a structural correction. Figure 3 visualizes the complexity of club-level market dynamics by

plotting the predicted trajectory for all clubs individually. This faceted view highlights how clubs diverge from the global trend. The vertical position of each chart confirms the initial hierarchy (e.g., random Intercepts), with elite clubs like Manchester City and Bayern Munich maintaining a consistently higher baseline than selling clubs (e.g., Sevilla, Porto). The distinct curvature of each line, particularly around the 2020 pandemic baseline, illustrates the significant variance in the random slopes. Clubs with concave trajectories (e.g., Juventus, Barcelona) plateaued or declined rapidly, confirming the risk of stagnation. Conversely, clubs with convex trajectories (e.g., Arsenal, Liverpool) show stronger momentum, indicating a more resilient recovery and active value generation throughout the decade.

Figure 3. Club-Specific Market Value Trajectories (2013–2024)



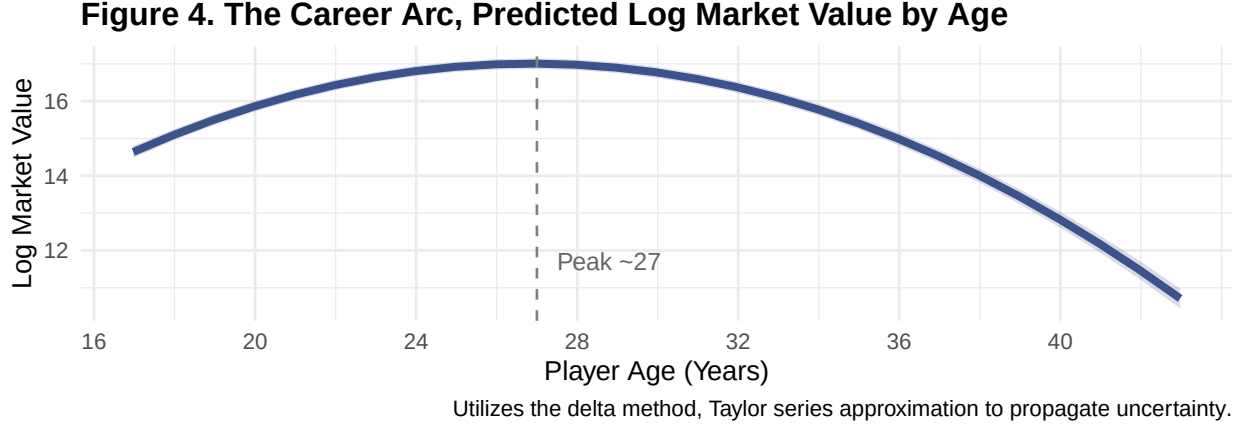
Dashed line marks pandemic onset; panels ordered by baseline prestige.

3.0.2 Biological Age

Our model predicts the peak age to be 26.9⁶ for optimal transfer market value we visualize in Figure 4 the biological constraints on valuation. The curve rises steeply as players transition

⁶We use the delta method, specifically Taylor series approximation, to propagate uncertainty through nonlinear transformation of $-\beta_{age}/(2 \times \beta_{age^2})$.

from ‘prospects’ (Age 17–23), flattens into a ‘prime’ plateau between ages 26 and 28, and declines steadily thereafter. The model predicts that a player’s value drops by approximately 1.5 log points (from a peak of 17.0 to 15.5) as they age from 27 to 35, reflecting the market’s heavy discount on older assets with limited resale potential.

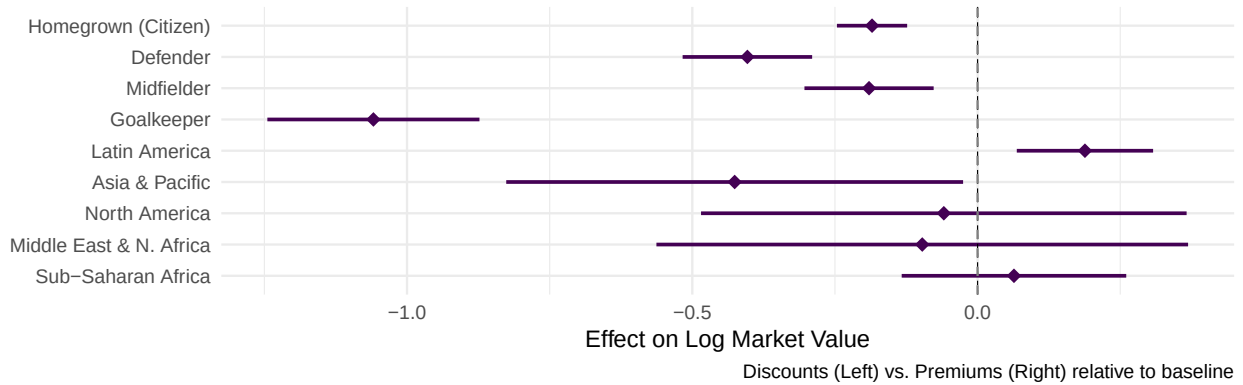


3.0.3 Contextual Premiums and Discounts

Squad status remains the dominant behavioral driver. The standardized coefficient for minutes played ($\pi_6 = 0.35$) indicates a 42% valuation premium ($e^{0.35} - 1$) for key starters (+1 SD) compared to average players. Conversely, we observe a robust “Domestic Discount,” where homegrown citizens are valued approximately 17% lower ($\gamma_2 = -0.19$) than comparable foreign imports. While we hypothesized a “Diversity Premium” for clubs with higher international representation, the fixed effect for `foreign_players` was positive but not statistically significant ($\pi_7 = 0.10, p = 0.32$) in the final model. Similarly, the `uefa_coefficient` showed no significant effect ($p = 0.70$). This suggests that once we account for the massive heterogeneity in Club Random Effects (i.e., Baseline Prestige and Growth Trajectories) and individual player reputation ($ICC \approx 0.78$), specific club-level metrics add little unique explanatory power. In Figure 5, we present relative market valuation for Level 2 fixed effects (e.g., citizenship, region origin, and position). The coefficient for “Homegrown (Citizen)” falls strictly to the left of zero, confirming that domestic players are valued significantly lower than comparable foreign imports ($\gamma_2 = -0.19$), potentially reflecting the recruitment

premium required to attract international talent. Attackers (the baseline) command the highest market value. All other positions show significant negative effects, with Goalkeepers facing the steepest penalty ($\gamma_{pos} = -1.06$), suggesting the market disproportionately rewards offensive output. While most regional coefficients overlap with zero (indicating no difference from the European baseline), Latin America stands out as the only region with a positive premium ($\gamma_{region} = 0.19$), suggesting South American players function as a “premium export brand” in the global market.

Figure 5. Valuation Gaps Across Citizenship, Position, and Region.



4 Discussion

This study demonstrates that player market value in elite football is driven primarily by persistent individual reputation ($ICC = 0.78$) but is meaningfully moderated by club context and market cycles. Our cross-classified approach properly attributes variance to portable player effects versus transient club effects.

Centering our temporal polynomial at 2020 provides direct evidence that the COVID-19 pandemic represented a fundamental discontinuity in the transfer market. The near-zero linear slope at this inflection point indicates that the decade-long inflationary trend abruptly halted, followed by market correction. The strict inverted-U relationship between age and market value, peaking at approximately 27 years, confirms that transfer markets price both

current ability and future option value. The steepness of the post-peak decline (approximately 1.5 log points from age 27 to 35) reflects forward-looking assessment of remaining productive years.

The dominance of minutes played ($\pi_6 = 0.35$) over position-specific metrics suggests that markets rely heavily on managerial assessments as signals of underlying quality, consistent with signaling theory. A one standard deviation increase in relative playing time yields a 42% valuation premium. The finding that homegrown players are valued approximately 17% lower than comparable foreign imports ($\gamma_2 = -0.19$) suggests domestic players may be systematically undervalued while international recruitment requires competitive bidding that inflates prices.

In conclusion, the substantial player-level variance confirms that persistent individual reputation dominates market assessments, while club context creates meaningful premiums and growth trajectories. The transfer market incorporates forward-looking assessments of biological decline, managerial signals through squad selection, and contextual factors that create systematic premiums and discounts. These patterns represent both economic regularities that clubs can anticipate and potential inefficiencies that astute market participants can exploit.

References

- Deloitte. (2021). *Riding the challenge: Annual Review of Football Finance 2021*. Deloitte Sports Business Group. <https://www2.deloitte.com/uk/en/pages/sports-business-group/articles/annual-review-of-football-finance-europe.html>
- Drewes, M., Daumann, F., & Follert, F. (2021). Exploring the sports economic impact of COVID-19 on professional soccer. *Soccer & Society*, 22(1-2), 125-137. <https://doi.org/10.1080/14660970.2020.1802256>
- Herm, S., Callsen-Bracker, H. M., & Kreis, H. (2014). When the crowd evaluates soccer players' market values: Accuracy and evaluation attributes of an online community. *Sport Management Review*, 17(4), 484-492. <https://doi.org/10.1016/j.smr.2013.12.006>
- Maas, C. J. M., & Hox, J. J. (2004). Robustness issues in multilevel regression analysis. *Statistica Neerlandica*, 58(2), 127-137. <https://doi.org/10.1046/j.0039-0402.2003.00252.x>
- Poli, R., Ravenel, L., & Besson, R. (2017). *Transfer market analysis* (CIES Football Observatory Monthly Report No. 27). CIES Football Observatory. <https://football-observatory.com/IMG/sites/mr/mr27/en/>

5 Appendix A

Table 1: Variable Descriptions and Descriptive Statistics ($N=7,168$ raw)

Description	Raw Stats (Mean/SD)	Transformation Strategy
Continuous Variables		
Market Value (2024 USD)	\$25.8M (29.0M)	<i>Log-transformed (ln)</i>
Season Year	2018.63 (3.47)	<i>Centered at 2020 (0 = 2020)</i>
Player Age (Years)	26.50 (4.58)	<i>Grand-mean centered + Quadratic</i>
Height (cm)	182.10 (6.84)	<i>Grand-mean centered</i>
Minutes Played	1761.77 (1337.70)	<i>Z-scored within Club-Year</i>
Goals Scored	3.41 (6.25)	<i>Centered by Position Mean</i>
Assists Made	2.79 (4.00)	<i>Centered by Position Mean</i>
Club Foreign %	63.5% (11.1%)	<i>Grand-mean centered</i>
UEFA Coefficient	20.42 (6.89)	<i>Year-mean centered</i>
Red Cards	0.07 (0.28)	<i>Raw count</i>
Yellow Cards	3.42 (3.60)	<i>Raw count</i>
Categorical Variables		
Citizenship (Homegrown)		<i>Binary (1=Citizen)</i>
- No	4498 (62.8%)	
- Yes	2567 (35.8%)	
- NA	103 (1.4%)	
Primary Position		<i>Factor (Ref: Attack)</i>
- Attack	1925 (26.9%)	
- Defender	2448 (34.2%)	
- Goalkeeper	605 (8.4%)	
- Midfield	2190 (30.6%)	
Region of Origin		<i>Factor (Ref: Europe)</i>
- East Asia & Pacific	69 (1.0%)	
- Europe & Central Asia	5139 (71.7%)	
- Latin America &	1441 (20.1%)	
Caribbean		
- Middle East & North	56 (0.8%)	
Africa		
- North America	52 (0.7%)	
- Sub-Saharan Africa	303 (4.2%)	
- NA	108 (1.5%)	

6 Appendix B

6.1 Annotated Model Syntax

```
# -----  
# MODEL SPECIFICATION & FITTING  
  
# 1. MODEL NULL (Unconditional Means)  
# Purpose: Partition variance (calculate ICC) between Player and Club levels  
model_null <- lmer(usd_revenue_2024_log ~ (1|player_id) + (1|club_id),  
                  # the following repeats for all models captured as "..."  
                  data = clean_player_df,  
                  REML = TRUE,  
                  control = strict_control)  
  
# 2. MODEL LINEAR TIME TREND  
# Purpose: Test for general linear inflation in market values  
## (Base model for time)  
model_time_linear <- lmer(usd_revenue_2024_log ~ season_year_c +  
                          (1|player_id) + (1|club_id), ...) # Random Effects  
  
# 3. MODEL CUBIC TIME TREND (Fixed Effects)  
# Purpose: Test for non-linear "Boom and Bust" cycles  
model_time_cubic <- lmer(usd_revenue_2024_log ~  
                          # Temporal Controls  
                          ## (raw=TRUE for interpretability)  
                          poly(season_year_c, 3, raw=TRUE) +
```

```

(1|player_id) + (1|club_id), ...)

# 4. MODEL RANDOM QUADRATIC SLOPE (Club Level)
# Purpose: Test if trajectory curvature varies by club
## (Linear slope model omitted for brevity)
model_slope_quadratic <- lmer(usd_revenue_2024_log ~
    poly(season_year_c, 3, raw=TRUE) +
    (1|player_id) +
    (poly(season_year_c, 2, raw=TRUE) | club_id),
    ...)

# 5. MODEL (FINAL): FULL SPECIFICATION
# Purpose: Estimate marginal effects of performance, status,
## and context controlling for time
model_player <- lmer(
    usd_revenue_2024_log ~
    poly(season_year_c, 3, raw=TRUE) +
    poly(age_c, 2, raw=TRUE) +                # Career Arc
    # Player context
    height_c + position + Region + citizen_of_club_country +
    # Performance
    goals + assists + red_cards + yellow_cards + minutes_played +
    uefa_coefficient + foreign_players +        # club fixed effects
    (1|player_id) + (poly(season_year_c, 2, raw=TRUE) | club_id),
    ...)

# -----

```



```

# LIKELIHOOD RATIO TEST (MIXTURE DISTRIBUTION)

# LRT test for 2 random slopes
calculate_lrt_mixture <- function(reduced_model, full_model) {

  # 1. Extract Log-Likelihoods
  ll_reduced = as.numeric(logLik(reduced_model))
  ll_full = as.numeric(logLik(full_model))

  # 2. Calculate LRT Statistic
  lrt_stat = -2 * (ll_reduced - ll_full)

  # 3. Calculate Difference in Parameters (df)
  df_reduced = attr(logLik(reduced_model), "df")
  df_full = attr(logLik(full_model), "df")
  df_diff = df_full - df_reduced

  # 4. Calculate P-Value based on the difference

  if (df_diff == 3) {
    # Going from 2 Random Effects (Int+Lin) -> 3 Random Effects Int+Lin+Quad
    # We add 1 variance + 2 covariances = 3 parameters.
    # The mixture is 0.5 * ChiSq(2) + 0.5 * ChiSq(3)
    p_val = 0.5 * pchisq(lrt_stat, df = 2, lower.tail = FALSE) +
      0.5 * pchisq(lrt_stat, df = 3, lower.tail = FALSE)

  } else if (df_diff == 2) {

```

```

# Going from 1 Random Effect (Int) -> 2 Random Effects (Int+Lin)
# Adds 1 variance + 1 covariance = 2 parameters
p_val = 0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE) +
  0.5 * pchisq(lrt_stat, df = 2, lower.tail = FALSE)

} else if (df_diff == 1) {
  # Adding a single parameter (e.g. just a variance, no covariance)
  p_val = 0.5 * pchisq(lrt_stat, df = 0, lower.tail = FALSE) +
    0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE)

} else {
  # Fallback for standard fixed effects comparisons
  p_val = pchisq(lrt_stat, df = df_diff, lower.tail = FALSE)
}

# Output
cat("LRT Statistic:", round(lrt_stat, 4), df_diff, "\n")
cat("P-Value:      ", format.pval(p_val, eps = .0001), "\n")
}

```

LRT Statistic: 97.1306

Parameter Diff: 5

Distribution: Standard ChiSq(5)

P-Value: < 0.0001

7 Appendix C

Table 1. Model Fit Comparison (AIC, BIC, and ICC)

Models	Purpose & Research Justification	ICCplayer/ club	AIC	BIC
Null	Calculate ICC and partition variance between players and clubs before adding predictors	.75/.03	20105.19	20132.62
Time Linear	Test for a general linear trend in market value growth	.76/.03	20050.17	20084.46
Time Cubic	Test for non-linear market dynamics (e.g., acceleration or deceleration/plateaus)	.76/.03	20021.61	20069.61
Random Slope Quadratic Varying Level 1 fixed effects	Test if the growth curvature varies by club	.76/.04	19934.48	20016.76
Final Model Selection	Estimate the marginal value of performance metrics and player/club characteristics controlling for time	.78/.05	14467.31	14679.87
	Compare models using LRT, AIC/BIC, and residual diagnostics (QQ plots) to select the most parsimonious and robust fit			

Figure 6. Model Diagnostic Plots

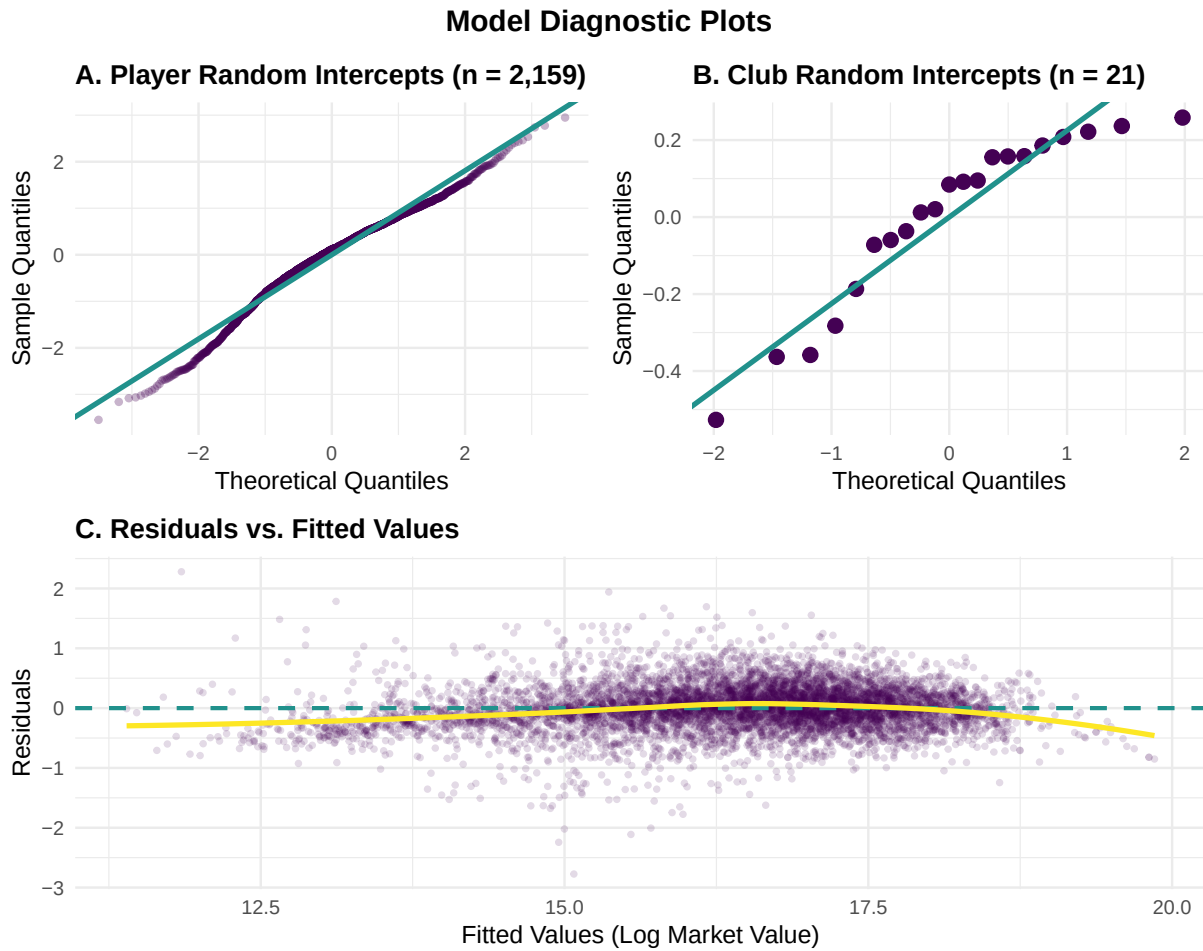


Figure 1: Diagnostic plots: (A) Q-Q plot for player random intercepts, (B) Q-Q plot for club random intercepts, (C) residuals versus fitted values.

Figure 7. Club Random Effects Correlation Structure

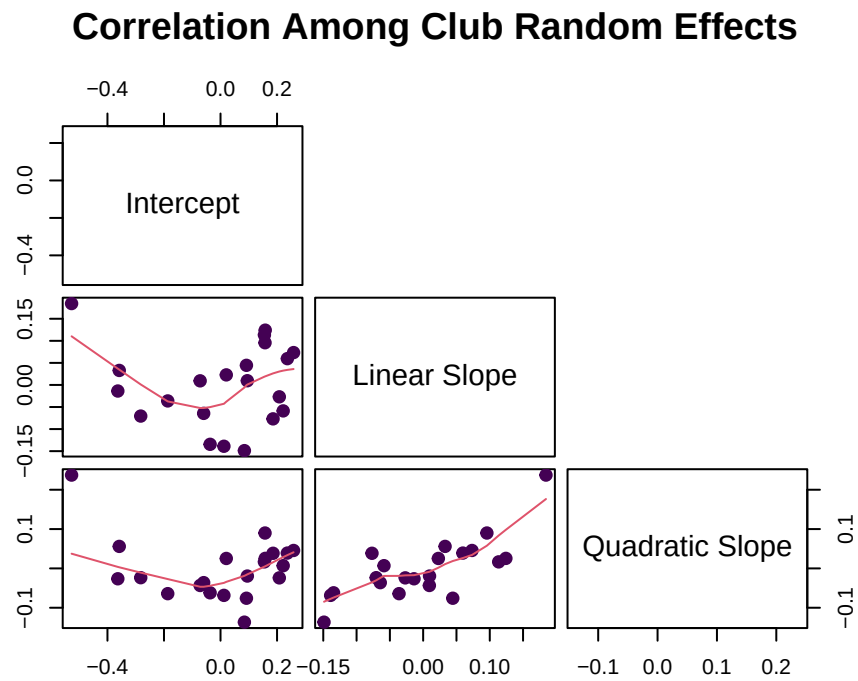


Figure 2: Correlations among club-level random effects (intercept, linear slope, quadratic slope).

8 Appendix D

Acknowledgement of AI Assistance

We acknowledge the use of AI-assisted tools (GitHub Copilot and Claude) during the preparation of this manuscript. These tools were used for:

1. **Coding assistance:** Generating and debugging R code for data visualization, particularly for creating the faceted club trajectory plots (Figure 3) and diagnostic visualizations (Figure 6).
2. **Resolving convergence issues:** Troubleshooting model convergence problems with the cross-classified multilevel model, including optimizer selection (BOBYQA) and parameter constraints.
3. **Code documentation:** Improving comments and annotations in the model syntax presented in Appendix B.

All statistical analyses, interpretations, and substantive conclusions were developed and verified by the authors. The AI tools served as coding assistants and did not contribute to the research design, data collection, or theoretical framework of this study.